## VLSM Magic Number Table

| Student id. | |
|---|---|

| Student Name .. | |
|---|---|
| Project .. | |

| Row | | Bits | No. of Addresses | VLAN Name | Req'd Hosts | Magic Num. | Subnet Address | Subnet Mask | | Cidr / Notation |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | | $2^{0}$ | 1 | | | | | | | |
| 1 | | $2^{1}$ | 2 | | | | | | | |
| 2 | | $2^{2}$ | 4 | | | 4 | | 255.255.255.252 | | /30 |
| 3 | | $2^{3}$ | 8 | | | 8 | | 255.255.255.248 | | /29 |
| 4 | | $2^{4}$ | 16 | | | 16 | | 255.255.255.240 | | /28 |
| 5 | | $2^{5}$ | 32 | | | 32 | | 255.255.255.224 | | /27 |
| 6 | | $2^{6}$ | 64 | | | 64 | | 255.255.255.192 | | /26 |
| 7 | | $2^{7}$ | 128 | | | 128 | | 255.255.255.128 | | /25 |
| 8 | | $2^{8}$ | 256 | | | 1 | | 255.255.255.0 | | /24 |
| 9 | | $2^{9}$ | 512 | | | 2 | | 255.255.254.0 | | /23 |
| 10 | | $2^{10}$ | 1024 | | | 4 | | 255.255.252.0 | | /22 |
| 11 | | $2^{11}$ | 2048 | | | 8 | | 255.255.248.0 | | /21 |
| 12 | | $2^{12}$ | 4096 | | | 16 | | 255.255.240.0 | | /20 |
| 13 | | $2^{13}$ | 8192 | | | 32 | | 255.255.224.0 | | /19 |
| 14 | | $2^{14}$ | 16384 | | | 64 | | 255.255.192.0 | | /18 |
| 15 | | $2^{15}$ | 32768 | | | 128 | | 255.255.128.0 | | /17 |
| 16 | | $2^{16}$ | 65536 | | | 1 | | 255.255.0.0 | | /16 |
| 17 | | $2^{17}$ | 131072 | | | 2 | | 255.254.0.0 | | /15 |